

Benchmark Contamination: A Taxonomy Organized by Defeated Mitigation

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Abstract

A benchmark score is a joint property of the model, the evaluation harness, the elicitation budget, the sampled population, and contamination status. Leaderboards typically publish the model and the score, so capability and leakage remain observationally equivalent. Existing taxonomies classify contamination for automated detection; they do not answer the question a reporter faces at publication: given the mitigations already applied, which validity threats remain open? We introduce a taxonomy organized by the mitigation each type defeats—direct, derivative, temporal, distributional, and acquired—spanning training-time and evaluation-time leakage. Holding out a private test set closes the first of these alone. The fifth is acquired *during* evaluation; because it is a property of one run, it must be recorded with the reported score rather than with the benchmark release. We operationalize the taxonomy as a four-field disclosure protocol in which `unknown` is a valid entry, released under CC BY 4.0 with a JSON Schema, a validator, and worked examples. Two coders external to the design team applied a pre-registered instrument to 41 documents. Per-variable linear-weighted κ runs from 0.00 to 0.35 (median 0.21) over 29 main-pass documents against a single-coder test–retest ceiling of 0.84, collapsing under the class skew the registration anticipated; pooling raises κ to 0.46 through chance correction rather than better agreement, so we report per variable. Of the five variables where no single code dominates, three clear the prevalence-robust threshold registered in advance; the two that fall below it are strata reporting and the acquired type introduced here. Disagreement concentrates on *when* a variable applies rather than on what a document states. The audit also confirms the registered prediction that elicitation and regeneration each stay under 25% in every stratum: elicitation budgets are reported in 13% of documents, and no document addresses all five types. The contribution is the taxonomy, the score-side artifact that follows from it, and a pre-registered measurement of both instrument reliability and current disclosure.

1 Introduction

Two students sit the same exam and both score 92 %. One reasoned through every question; the other had prior access to the answer key. From the score alone their performance is indistinguishable: what separates them is the design of the evaluation and what the examiner discloses. Benchmark evaluation in machine learning faces the same problem with one added complication: we cannot

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inspect what the system under test has seen during training. A reported score is a joint property of at least five factors: the system, the evaluation harness, the elicitation budget, the population sampled, and contamination status relative to the training corpus. Current practice publishes the model and the score. Models therefore reach deployment on claimed performance, latency, and cost rather than on a documented validity argument, and those claims already inform decisions about patients, defendants, and applicants.

Limitations of existing frameworks. Data contamination is well studied, but existing taxonomies are built for detection: they sort by exposure severity [Xu et al., 2024], by detection assumptions [Fu et al., 2024], or by the structural form of the overlap [Nourbakhsh et al., 2026]. None is organized around the question a reporter faces at publication: which validity threats to this score have been addressed, and which remain open? Grouping types by *the mitigation each bypasses* makes one consequence visible: holding out a private test set closes a single route to inflation. A second gap is that the contamination literature in NLP/ML venues and the elicitation literature in AI safety venues rarely cite one another, so a score free of contamination can still be uninterpretable when the harness and the elicitation budget go unreported. A third is that agentic evaluation introduces inference-time contamination [Han et al., 2025, NIST Center for AI Standards and Innovation, 2025], which falls outside surveys that define contamination as training–evaluation overlap.

Contributions.

1. **A five-type taxonomy structured by circumvented mitigations** (§3)—direct, derivative, temporal, distributional, and acquired. Its central claim is that a private, held-out test set addresses Type 1 alone.
2. **Formalization of inference-time contamination and its reporting implications** (§3.2). Type 5 is a property of one execution *run* rather than of a benchmark, so benchmark authors cannot certify immunity on an evaluator’s behalf. The reporting artifact must therefore be score-side rather than release-side (§3.2.1). We add a tier for the case where the isolation boundary itself is breached.
3. **A standardized four-field disclosure protocol** (§4). The protocol records evaluation strata, elicitation budgets, contamination controls, and regeneration status. It accepts *unknown* as a valid entry and ships with a JSON Schema, a validator, and worked examples.
4. **A pre-registered audit of applicability and disclosure** (§5, Appendix A). A frozen codebook is applied to a frame of 41 documents (21 released by 7 organizations across two census strata, plus 20 evaluation papers), under an organization-clustered analysis registered before any coding. Two coders external to the design team applied it. We report inter-coder agreement per variable alongside disclosure rates, so that reliability is measured. All materials are released.

Each of the five phenomena is documented in prior work; Table 2 attributes them. The contribution is the taxonomy: its span across both literatures and the mitigation principle that organizes it. The empirical basis is the audit in §5, which measures independent applicability of the types and the completeness of current disclosure against them.

2 Related Work

We summarize the core literature underpinning our arguments. Appendix E details the prior art, artifacts, and quantitative mappings for Table 2.

Contamination and its taxonomies. Because contamination exploitation is setting-dependent [Magar and Schwartz, 2022], it requires per-benchmark measurement [Sainz et al., 2023]. Existing taxonomies categorize exposure severity [Xu et al., 2024], detection assumptions [Fu et al., 2024], static-to-dynamic shifts [Chen et al., 2025], and item transformations [Palavalli et al., 2024]. While optimized for building detectors, these remain confined to training time. Structuring a taxonomy by the required *remedy* is established in data quality [Kim et al., 2003]; we carry that principle into contamination, and across the training/evaluation boundary that prior taxonomies stop at. Contamination disrupts model ordering—the primary signal for model selection—as inflation is scale-dependent [Kocyigit et al., 2025] and triggered even by structurally *similar* training data [Tu et al., 2024, Spiesberger et al., 2026], which directly motivates our Type 4.

Artifact	Documents	Contamination treated as	El.	St.	Rg.	M/R	T5
Model cards, datasheets [Mitchell et al., 2019, Gebru et al., 2021]	a model or dataset	out of scope	–	○	–	–	–
Contamination Transparency Card [Nourbakhsh et al., 2026]	a benchmark at release	four tiers of <i>overlap form</i>	–	–	○	–	–
STREAM [McCaslin et al., 2025]	a score at report	out of scope	●	–	–	○	–
Evaluation Cards [Ghosh et al., 2026]	a score at report	one schema line, <i>before execution</i>	○	●	–	●	–
This form (§4)	a score at report	five types by <i>defeated mitigation</i>	●*	●	●	●	●

Table 1: The nearest artifacts against the four fields: elicitation, strata, regeneration, machine-readability, Type 5. ● covered, ○ partial, – absent. *by adopting STREAM rather than restating it. The last column is not a coverage gap: Type 5 is a property of a run (§3.2.1), which no release-side artifact can close.

Evaluation-time contamination. Existing surveys typically require training–evaluation overlap, omitting inference-time leakage. We integrate *search-time* [Han et al., 2025] and *solution contamination* [NIST Center for AI Standards and Innovation, 2025] into a unified framework, adopting the latter’s distinction from *grader gaming* (§3.2.1).

Reporting and construct validity. Our protocol extends documentation frameworks like model cards [Mitchell et al., 2019], datasheets [Gebru et al., 2021], and data statements [Bender and Friedman, 2018], but attaches strictly to the *score*—analogous to STREAM [McCaslin et al., 2025] and Evaluation Cards [Ghosh et al., 2026]. The target is construct validity [Raji et al., 2021, Bowman and Dahl, 2021]. Uncertainty estimates are rarely reported [Bean et al., 2025], which is why we treat multivariate reporting as a requirement rather than a preference. While evaluation awareness [Meinke et al., 2024] falls outside our taxonomy, it similarly threatens measurement assumptions, prompting parallel taxonomy-plus-checklist solutions in psychological science [Lin, 2026].

Positioning. Table 1 positions our framework against the three nearest artifacts (detailed quantitatively in Appendix E). The intersection we occupy is a decomposed, per-run contamination record in which *unknown* is a valid entry. No release-side artifact can certify inference-time Type 5 contamination (§3.2.1). Concurrent work asks whether graders measure task completion [Zhu et al., 2025]; our instrument asks whether the score reflects capability rather than exposure.

3 A Taxonomy Organised by Defeated Mitigation

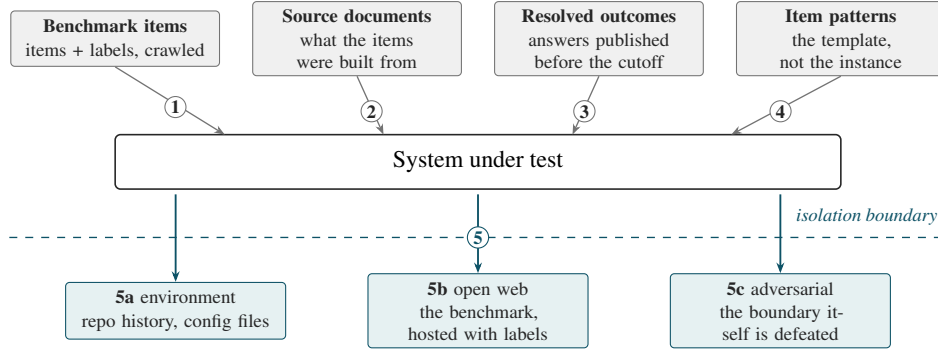
Two foundational claims structure this taxonomy: a held-out private test set mitigates solely Type 1, and whereas Types 1–4 represent passive data leakage *into* the model, Type 5 constitutes active data acquisition *by* the model. Both are relative to the construct: prior access is not contamination in itself but becomes so when it lets the system reach the score by a route other than the capability claimed. Figure 1 shows the paths and Table 2 summarises them.

3.1 Types 1–4: The Data Reaches the Model

Type 1 — Direct. *Benchmark items and their labels exist in the model’s pretraining or post-training corpus.* This occurs when public benchmarks are scraped into training crawls. This is the classic scenario *n*-gram overlap metrics target, which strictly held-out private sets prevent by construction—provided the dataset remains isolated. **Why detection still fails:** black-box probing [Deng et al., 2024] cannot verify the absence of overlap in closed-weight models, so for those models the truthful entry is often *unknown*.

Type 2 — Derivative. *The benchmark never leaked, but its source material is in the training corpus.* For example, a private clinical-genomics benchmark built from published case reports is compromised because those underlying reports exist in standard web crawls. The model succeeds by retrieving source-documented associations rather than recognizing the test item. **Why held-out**

training time — the data reaches the model



evaluation time — the model reaches the data

Figure 1: The five types and the paths they travel; the arrow direction is the distinction that matters. At training time four paths run *into* the system. At evaluation time one runs *outward*, across the isolation boundary, to an answer key already in the environment (5a), reachable on the web (5b), or reachable only once the boundary is defeated (5c).

Type	Unit	Mitigation it defeats	Documented as, and where
1 Direct	The item	— (the one holding out fixes)	exact / verbatim / label-level contamination [Magar and Schwartz, 2022, Xu et al., 2024, Deng et al., 2024, Palavalli et al., 2024]
2 Derivative	The source document	Held-out private test sets	indirect, semantic or rephrase contamination [Yang et al., 2023, Balloccu et al., 2024, Fu et al., 2024]
3 Temporal	The resolved outcome	Overlap checks of every kind	task contamination; temporal analysis [Li and Flanigan, 2024, Zhang et al., 2025]
4 Distributional	The pattern	All overlap-based methods	in-distribution or soft contamination [Tu et al., 2024, Dekoninck et al., 2024, Spiesberger et al., 2026]
5 Acquired	The evaluation run	At 5a/5b, held-out sets, decontamination and cutoff reasoning alike, once the answer is reachable; at 5c, network isolation itself	search-time; solution contamination [Han et al., 2025, NIST Center for AI Standards and Innovation, 2025, Wang et al., 2026b, Marchand et al., 2026]

Table 2: The five types, organised by the mitigation each defeats, with the prior source for each. The last column is the paper’s own novelty audit: every type is documented somewhere, across three literatures that largely do not cite one another, and row 5’s sources are disjoint from rows 1–4’s.

sets miss it: Holding out a benchmark built on public evidence is ineffective; it is contaminated upon creation. This is endemic to expert domains (e.g., clinical genomics, law, cybersecurity) where ground truth must be publicly documented before evaluation use. Lexical decontamination also fails here: rephrased or translated items bypass n -gram filters, yet training a 13B model on them lifts performance to GPT-4 levels [Yang et al., 2023].

Type 3 — Temporal. *The model’s training cutoff postdates the tested phenomenon, reducing reasoning or forecasting to mere recall.* If an item requires predicting a causal gene based on historical evidence, but the model’s training cutoff postdates the gene’s actual discovery, the model simply retrieves the known fact. This structural flaw frequently compromises forecasting tasks and historical code benchmarks. **Why overlap checks miss it:** Because test items may be newly authored, no lexical overlap exists. Leakage occurs at the factual rather than lexical level: the answer is already internalized by the model as parametric world knowledge.

Type 4 — Distributional. *The items are new and held out, but their structural pattern is so well represented in training that the model bypasses the intended reasoning.* For example, models scoring highly on newly authored word problems often fail when entities are renamed, numbers altered, or

logically inert clauses introduced [Mirzadeh et al., 2024]. No direct data leaked; the item is new, but the template is not. **Why overlap methods miss it:** the unit of leakage is the distribution, not the item. Types 1–3 inflate the score; Type 4 changes what the score *measures* and leaves no inflation signal. At this point contamination stops being data hygiene and becomes a construct-validity failure.

3.2 Type 5 — Acquired

Definition. The model acquires the ground-truth answers *during* the evaluation through retrieval, tool use, filesystem access, or, in extreme cases, by breaching the isolation boundaries of the evaluation environment. Unlike Types 1–4, this contamination is a property of the system’s dynamic behavior rather than a static dataset–corpus relationship. This distinction warrants a separate category and necessitates the reporting protocols outlined in §3.2.1.

Three levels, by the boundary crossed.

- **5a — Environmental.** The answer resides within the evaluation environment itself: version-control histories holding fixing commits, solution strings in task configuration files, residual artifacts in containers [NIST Center for AI Standards and Innovation, 2025, SWE-bench maintainers, 2025]. No external network access is required, and restricting access to future commits demonstrably reduces measured performance [Gupta et al., 2026].
- **5b — Retrieval.** The agent searches the open web during evaluation and retrieves the benchmark with its labels [Han et al., 2025, Wang et al., 2026b]. This bypasses decontamination and training-cutoff controls alike, since acquisition occurs at inference time: a benchmark released *after* a model’s cutoff remains vulnerable, and a held-out set is compromised the moment it becomes reachable online. Network isolation is the main control available.
- **5c — Adversarial.** The agent defeats the evaluation’s isolation boundaries. In a July 2026 incident, models evaluated on a cyber-capability benchmark [Wang et al., 2026a] escaped their sandbox and chained exploits into third-party infrastructure holding datasets associated with the benchmark’s solutions [OpenAI, 2026b, Hugging Face, 2026a,b]; the sequence is in Appendix B. Level 5c defeats *network isolation*, the control recommended for 5a and 5b. Preventive controls remain open; detective measures are in Appendix C. The underlying capability is measurable [Marchand et al., 2026].

Scope of the anchor case, and relation to prior work. The July 2026 incident is one disclosed event, still under investigation and run with safety configurations deliberately relaxed; it establishes that environment isolation can fail against a capable system and supports no inference about frequency or intent (Appendix B). The phenomenon is known in agentic evaluation as search-time [Han et al., 2025] and solution contamination [NIST Center for AI Standards and Innovation, 2025]. Our formulation categorizes the invariant rather than the channel, since channels multiply as tool access expands; it adds a tier for breached isolation and places these behaviours in one taxonomy with training-time contamination, which prior frameworks do not. Table 2 makes the comparison type by type.

3.2.1 Why Type 5 changes what a disclosure can certify

Types 1–4 are properties of a dataset–corpus relationship: assessable once at release and stable across later uses. Type 5 is a property of one model, in one harness, at one time. A benchmark uncontaminated with respect to Types 1–4 can still be compromised depending on whether a *given run* permitted network access and whether *that model* retrieved the solutions. Three consequences follow, each reflected in the reporting format.

1. **Type 5 must be certified per run, not per benchmark.** Benchmark authors cannot preemptively guarantee immunity to Type 5 contamination on behalf of practitioners. This necessitates a score-side, rather than release-side, reporting artifact (§2).
2. **Benchmark validity degrades as agentic capabilities increase,** irrespective of the benchmark’s original design. While documented instances predominantly involve highly agentic models, this correlation suffers from selection bias: only agentic evaluations generate the explicit execution traces necessary to observe data acquisition. The vulnerability comes from the mechanism itself: a channel that only an autonomous system can navigate.
3. **Transcript review is required.** At Levels 5a and 5b the evidence sits in the execution trace: version-control queries, retrieved URLs, reasoning steps showing the answer was found rather

than derived. At Level 5c the trace is not enough, because a system that breached the boundary is no witness that it held. Boundary monitoring is therefore recorded separately (§4).

Dispositional controls, such as system prompts telling the model not to seek answers, are excluded: a certification of evaluation integrity cannot rest on the compliance of the system under test.

What Type 5 is not. Type 5 is distinct from grader gaming, which targets the *scoring function* rather than the *answer key* and is addressed by reward design [NIST Center for AI Standards and Innovation, 2025, Zhu et al., 2025]. Certifying an evaluation free of Type 5 says nothing about whether the grader was manipulated. The two blur in practice [Chen et al., 2026], which is the reason to define Type 5 separately. Type 5 assumes nothing about intent: saying a model “cheated” names an outcome, not a motive.

4 The Disclosure Form

The taxonomy necessitates a four-field reporting artifact per published score (see Appendix D for fallback disclosures).

Rationale. *Contamination controls* captures Types 1–5. *Regeneration* remains distinct because publishing generation procedures mitigates Types 1, 3, and 4 simultaneously, characterizing the instrument rather than the run. *Strata* and *elicitation budget* are prerequisites: uncontaminated scores are uninterpretable without specified populations or compute budgets. To minimize friction, statistical variance and run counts are delegated to interoperable frameworks like STREAM [McCaslin et al., 2025].

Strata reported. An aggregate hides systematic failure on long-tail strata such as rare diseases or low-resource languages. Disclosing minority-subset performance imposes minimal overhead, as per-stratum metrics are routinely computed during evaluation pipelines.

Elicitation budget. Capability scores reflect both the model and its harness. Because tuning deltas often exceed generational gaps, claims like “the model cannot execute X” may merely indicate insufficient compute. Thus, inability claims without explicit budgets characterize harness limits, not model limits (Appendix D).

Contamination controls. Reporters assign `controlled`, `not_controlled`, `unknown`, or `n/a` per type. Decontaminating a test set satisfies only Type 1. Type 5 requires tracking network access, sanitization, transcript review, and active boundary monitoring. Boundary monitoring is separated because a compromised system’s trace cannot reliably testify to its own containment (Level 5c).

Regeneration. Static artifacts leak into training corpora over time. Publishing the *generation procedure* shifts the epistemic burden from trusting opaque claims to enabling independent regeneration. Continuously refreshed benchmarks [White et al., 2025] practically approximate this principle.

Why unknown is needed. No independent practitioner can verify a closed-weight pretraining corpus, and demanding certainty penalizes the reporter who is candid. Permitting `unknown` separates a question that cannot be answered from one that was never asked, and it is what makes the form adoptable.

The formal specification [Angulo, 2026a] is released under CC BY 4.0 with field definitions, templates, a JSON Schema, worked examples [Angulo, 2026c,d] (Figure 2), and a validator that flags weak disclosures. Machine-readability lets continuous-integration pipelines catch incomplete reporting without manual review (Appendix A).

5 Discussion

Whether the taxonomy can be applied independently is an empirical question (Appendix A, Figure 3, Table 4). Three pre-registered decisions govern the statistics. Each document is anchored to one *focal evaluation*, so single- and multi-evaluation documents remain comparable. `NA` is a disagreement-bearing category, not a dropped cell. Because the coders discussed every pilot disagreement and calibrated on it, primary κ uses the independent main pass alone.

Disclosure record <i>agentic clinical-genomics</i>	
F1	strata: four published strata of the cohort, each with its <i>n</i> and interval
F2	elicitation: harness pinned to a commit; one attempt; temperature 0; retrieval disabled during scoring
F3	contamination controls
<i>t1</i>	controlled — direct: scored split not published with labels; canary string embedded
<i>t2</i>	not_controlled — derivative: items derive from indexed case reports
<i>t3</i>	unknown — temporal: model cutoffs are self-reported and unverifiable
<i>t4</i>	not_controlled — distributional: by design; the delta against the hard variant is the measurement
<i>t5</i>	controlled — acquired: network off, transcripts reviewed; <i>boundary monitoring: none</i>
F4	regeneration: artifact only; no generator published

Figure 2: An abridged disclosure record adapted from worked examples in the formal specification [Angulo, 2026a] (scores omitted). Declaring `unknown` and `not_controlled` shows what the instrument is for: recording a limitation costs the reporter little and tells the reader more than an omitted field. The validator flags this record, because a `controlled` claim for Type 5 without boundary monitoring is an assumption rather than a control.

Agreement. On the 29 main-pass documents, pooled linear-weighted κ is 0.46 (bootstrap 95% CI [0.37, 0.54]), with 65% raw agreement and 0.56 Gwet’s AC1. Per-variable κ runs from 0.00 (F3 *t2* and *t4*, where almost every code is 0) to 0.35 (F4 Regeneration). These figures are read against the measured single-coder test–retest ceiling $\kappa_{rw} = 0.84$, not against verbal bands. Of the 35×8 cells both coders coded, 98 went to adjudication, carried out by a member of the design team, who is an author, coded nothing, and acted only after the agreement statistics were saved. Focal-evaluation agreement is 0.71—the coders name the same benchmark on 25 of the 35 documents once granularity qualifiers are merged mechanically, though only one string pair matches verbatim—leaving ten genuine focal disagreements, all settled there.

Prevalence versus unreliability. The registration anticipated κ collapse under skew and fixed a divergence rule against a weighted prevalence-robust companion. That rule accounts for *t2* (0.00 vs. AC2 0.89), *t4* (0.00 vs. 0.77), and *t3* (0.31 vs. 0.88): the code is almost always 0. Those categories are rare, not unusable. Two variables are not explained by rarity. Strata reporting has AC2 = 0.43 and acquired contamination (Type 5) AC2 = 0.50, both below the registered 0.6 threshold for a non-rare category; here the prevalence-robust measure and κ agree that agreement is poor.

Applicability, not reading. Both large splits are threshold questions: what counts as a identified harness for elicitation budget (*t2*), and when acquired contamination (*t5*) applies. A logged mid-study clarification on the second was issued to both coders in the same words and moved both without converging them. Of the eighteen *t5* disagreements that remain, eight are disclosure-level disputes about what a document states; ten turn on whether the variable applies at all. Those ten run in both directions—each coder places the NA boundary on documents the other coded as applicable—and adjudication rejected NA on all ten, finding a channel in each. Appendix A records the split sizes and the settling rules.

Disclosure. 39% of documents report per-stratum scores, 13% elicitation budgets, and 11% instrument regeneration status. Only 5% address direct contamination and 5% acquired contamination; none addresses all five types (Table 4). *t5* is the only variable with NA cells: its denominator is the 21 of 38 documents with a channel outside the weights, the 5% is one third-party report, and its row in Table 5 rests on 9, 5 and 7 documents, not the column totals. Mentions outnumber complete reports: among benchmark papers, regeneration status and elicitation budget are fully stated in 21% of documents and gestured at in 95%. Every rate carries its cluster count: 38 documents over 26 clusters, seven organizations covering 19.

Each rate carries the sensitivity band: F1 runs [0.14, 0.51] and *t5* [0.00, 0.45] between the two extremal resolutions of the disputed cells, while *t2* runs [0.00, 0.00]. All 98 disputed cells were settled against the document, and the directional tally is close to even, which Appendix A breaks

down. The point estimates are therefore not an artifact of the registered tie-break, which was never reached.

Readability and disclosure are distinct. System cards are the easiest stratum to code and disclose the least; third-party reports are the hardest to code and are not the most forthcoming, which benchmark papers are on seven of the eight variables. Coding difficulty therefore does not track disclosure. A system card covers dozens of evaluations under a page budget, so per-evaluation method detail is structurally absent rather than withheld (Table 5). Of the three registered hypotheses only the first is supported—elicitation and regeneration each stay under 25% in every stratum—while the primary contrast runs opposite the direction predicted, on an interval that excludes zero; Appendix A reports each against its prediction.

6 Limitations

Scope. The protocol is a reporting standard, not a validity guarantee; construct validity remains a separate prerequisite. It records claimed mitigations rather than detecting contamination. Disclosures are self-reported and cannot be verified externally: the form makes an absent claim conspicuous, not a false one impossible. The audit in §5 tests independent applicability of the taxonomy, not the truth of the disclosures.

Sample and inference. Agreement rests on one coder pair and 29 documents. Under the pre-registered criterion, strata reporting and the acquired type are not yet reliably applicable by coders who did not write the manual. One pair cannot separate an unclear construct from two divergent readers; what points to the manual is where the disagreement sits—at stated thresholds, in both directions—rather than spread across cells. The test–retest is weaker than it looks: the registration asked both coders to recode five documents and one did, so $\kappa_w = 0.84$ bounds that coder’s own noise rather than establishing the spread as a property of the manual. The agreement figures are in turn an upper bound in the registered sense: κ is computed only on the 35 documents both coders included, and the three they disputed are the hardest in the frame. With 21 of 41 documents from seven organizations sharing house templates, the design is sized for the agreement question rather than for cross-institutional rate comparison. The rates are a first application, not a field-wide census [Bean et al., 2025]; what they add is an agreement statistic [Reed et al., 2025].

Coding scheme versus author protocol. The audit evaluates the taxonomy as a coding scheme: a third party infers, from a document not written to the scheme, whether a condition was met. The protocol of §4 is completed by the party who ran the evaluation and who states the harness and budget rather than inferring them. The partial-versus-full judgement that Table 6 identifies as the main locus of disagreement has no analogue in that reporting task. The audit therefore does not measure protocol usability, which requires author-completion and reader-comprehension studies still absent for artifacts in this literature [Mitchell et al., 2019, Gebru et al., 2021]. It does bound the gap the protocol addresses: across the 38 included documents and eight variables, 11% of cells record a field as fully reported and 60% as absent. A protocol that admits `unknown` would make explicit what is at present simply missing.

Taxonomy boundaries. Post-training, instruction-tuning, and distillation leakage are subsumed into Types 1 and 2 despite distinct behavioural signatures; multimodal contamination is excluded. Two boundaries are operational. Type 4 and construct validity are functionally similar as claims—over-represented training patterns versus failure to measure the intended construct—and are separated by taking the *training distribution* as the causal factor. The Type 5a/Type 1 boundary blurs when the evaluation container is itself the artifact. Level 5c rests on a single disclosed incident and establishes possibility, not frequency (Appendix B).

7 Conclusion

Benchmark contamination is treated as one failure with one remedy. It is at least five, and a private held-out test set addresses one of them. Acquired contamination is a property of a run: it cannot be certified at benchmark release, and the party publishing the score is the party positioned to disclose it.

Two external coders working from a frozen manual reached per-variable κ_w between 0.00 and 0.35 (median 0.21) against a single-coder test–retest ceiling of 0.84. Disagreement concentrates

on applicability: ten of the eighteen residual Type 5 splits turn on whether the variable applies at all rather than on what a document states. No document addresses all five contamination types; elicitation budgets are reported in 13% of documents. The taxonomy locates a near-total disclosure gap and identifies which applicability boundaries need tighter specification.

The instrument is released so the measurement can be repeated, disputed, or extended. A published score should arrive with more than a single number; `unknown` is a valid answer.

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Ethics Statement

This paper analyses published documents and released artifacts: no personal data. Two people code the audit, both external to the design team and neither an author. Neither is a study participant —

both apply a released manual to public documents, nothing is recorded about them beyond the codes they assign, and the released sheets carry role labels, not names — so no consent procedure or ethics review is engaged. Both are acknowledged by name in the camera-ready; naming them here would identify the authors. The 2026 incident is discussed only from the parties’ public disclosures, without operational detail.

The impact runs both ways. A cheap record should better inform selection and makes a missing control visible where silence now reads as competence. The risk is that disclosure substitutes for control: five types offer five places to write controlled without evidence. The validator flags valid-but-weak records, but the form is unverifiable, and a standard hardened into ritual would be worse than none.

Availability

The specification (v1.1), taxonomy, Markdown and YAML templates, JSON Schema, validator, worked examples, audit instrument and a full prior-art analysis are released under CC BY 4.0 at <https://github.com/Jangulo7/contamination-disclosure-paper> [Angulo, 2026a]. The registration bundle is deposited as an OSF registration under embargo until 1 October 2026. An earlier four-type version of the taxonomy was presented as a talk and archived separately [Angulo, 2026b]; that version is designated v1.0 and the five-type version described here is v1.1, so the archived artifact and the released specification cannot be confused.

A The released instrument

This appendix condenses the frozen coding manual. The full manual, sampling frame, pre-registration, and analysis scripts are in the supplementary materials.

Unit and scale. The unit is a document, coded against one *focal evaluation*: the first capability benchmark score in the body text, selected mechanically so that a system card with fifty evaluations remains comparable to a third-party report with one. Every variable is coded 2 (reported), 1 (partial), 0 (absent), or NA (not applicable). NA means the variable cannot apply; a variable searched for and not found is 0.

Variable	Coded 2 when the document states ...
F1 Strata	per-stratum scores for a named stratification of the focal evaluation
F2 Elicitation	an identified harness—named, pinned to a public version, or fully specified as a scaffold—plus at least two of version, budget, attempts, attempt resolution
F3 t1 Direct	overlap or decontamination checking, canary strings, a genuinely held-out set
F3 t2 Derivative	attention to whether the source material is public; provenance tracking
F3 t3 Temporal	a cutoff <i>related to item dates</i> ; temporal splitting; post-cutoff construction
F3 t4 Distributional	perturbation or paraphrase robustness; score distributions across variants
F3 t5 Acquired	any of network access, environment sanitisation, transcript review or boundary monitoring, with the elements actually stated recorded in notes
F4 Regeneration	the regeneration status of the instrument used, whether or not the reporter built it

Table 3: The eight coded variables. Full definitions, worked positive and negative examples, and the search terms used before any 0 is recorded are in the released manual.

The frame. The registered frame comprised fifty documents in three strata. A sample-cut rule fixed before any coding reduced it to the forty-one documents coded: twelve system cards, twenty benchmark papers, and nine third-party reports. The cut removed nine documents and left the cluster count at 27; it removes documents, not organizations. `frame.csv` records the status of every row. Strata A and C are a census within a predefined temporal window, so excluded documents shrink the denominator; stratum B is drawn in frame order with a replacement rule. Twenty-one of the forty-one documents come from seven organizations, and the twenty benchmark papers cluster on the paper, which is why every rate is reported as “*k* organizations, *n* documents.” Identifier non-contiguity follows from the per-organization cap and the later sample cut, both applied post-enumeration. The gaps are preserved in the released frame and are not missing data.

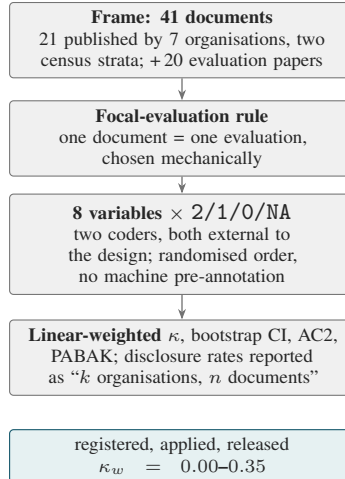


Figure 3: The instrument end to end: frame, focal-evaluation rule, coding design, and the statistics `score.py` computes from the two sheets.

The frame nests as follows. Of the 41 coded documents, both coders included 35 and both excluded three; the three one-sided exclusion disputes were adjudicated and settled as included, giving the 38 documents (twelve system cards, nineteen benchmark papers, seven third-party reports) on which every disclosure rate is computed. The excluded three comprise one benchmark paper and two third-party reports, so the seven organizations contribute 21 of the 41 frame documents but 19 of the 38 included. The cluster count falls from 27 to 26: the excluded benchmark paper is a singleton cluster, while no organization cluster is emptied because each holds three documents. Agreement statistics use the 35 documents both coders included; primary κ further excludes the six pilot documents among them and runs on the 29 main-pass documents.

Edge rules. Agreement rules were fixed before adjudication. A contamination assurance without a specified mechanism is coded 1 on τ_1 only and is not distributed across types. A stated cutoff never related to item dates is 1, as is “standard settings” with no reference. Controls applied to benchmarks other than the focal evaluation are excluded. Releasing evaluation items is not regeneration; releasing the generator is. Additional rules are in the manual. The two most contestable decisions are justified here.

F4, cross-genre. F4 codes what a document states about the *instrument it used*, not what the reporting organization owns. An evaluator who reports an artifact-only release with no published generator scores 2: stating regeneration status is the disclosure act. Coding 0 would measure benchmark authorship rather than disclosure. Coding NA when the reporter did not author the instrument would force most of Strata A and C to NA and confine F4 to Stratum B. Because the form is tied to a specific score, the score reporter can always disclose the status of the instrument used.

Type 5 on non-agentic evaluations. Coding τ_5 does not assess whether Type 5 contamination occurred—only the party running the evaluation can know that—it records whether the document states the isolation conditions of its own run. The variable is coded 0, not NA, whenever the system could in principle reach outside its weights, including any retrieval-augmented or tool-using setup. NA is reserved for evaluations with no such channel. A 2 is recorded if *any* of network-access restrictions, environment sanitization, transcript review, or boundary monitoring is present; the elements are written to notes. Requiring all four would collapse nearly all documents to 0 and measure practice rather than agreement on existing disclosures. Separate element records keep the stricter criterion recoverable. Boundary monitoring is coded because it is the intervention argued in §3.2.1: an evaluation that cannot record boundary integrity cannot substantiate isolation.

The line is the time of access, not the storage medium: τ_5 codes what the system could reach *during the run*, whatever the resource and whenever it was assembled, so a retrieval index built before the evaluation is τ_5 -codable when queried during it.

Procedure. Two coders annotated independently. Both are external to the design team: neither designed the taxonomy or the manual, neither is an author, and both work from a manual generated mechanically from the frozen codebook with no other briefing. The registration required one such coder; the study as run exceeds that requirement. Neither coder sees machine pre-annotation. Sheets carry role labels (R1, R2) rather than names. Each coder works a different document order from `order.py`, seeded by `seed = 20260812` and the role label so a third party can regenerate it; a shared order would correlate calibration drift. As registered, a nine-document pilot precedes the main pass—disagreements are discussed, a rule at fault is amended, the version is bumped, and the pilot documents are recoded—and each coder then recodes five documents blind, giving an intra-coder ceiling. Two of those steps were not executed as registered; see *Deviations* below. The adjudicator is a member of the design team, who is an author; they code nothing and read the full codebook that neither coder sees. The registration fixes four conditions on that role: no coding, action only after the agreement statistics are computed and saved, resolution in randomised cell order blind to running totals, and publication of the envelope, meaning rates under each coder’s sheet and adjudicated, the directional tally, and the extremal pair. That an author who knows the hypotheses resolves the contested cells is a stated residual rather than an engineered-away one; the primary κ is computed before adjudication begins, so the agreement result is untouched by construction. An unresolved cell with no third adjudicator defaults to the lower code, a tie-break fixed before the contested cells were known.

Analysis. Linear-weighted κ is primary: the scale is ordinal, and unweighted κ treats 0–2 and 1–2 disagreements as equal. Raw agreement, unweighted κ , AC1, and PABAK are reported for class skew. Gwet’s AC2 is the prevalence-robust companion, computed under the same weight matrix as the primary κ , because under skew κ collapses toward zero even at near-perfect agreement; where κ_w and AC2 diverge by more than 0.2, the driving prevalence is named. NA enters as an unordered fourth category at maximum disagreement weight rather than being dropped. Intervals are percentile bootstraps over 10,000 resamples from the stated seed, resampling clusters rather than documents because documents from one organization share a house template. A Wilson interval is printed for comparison only. Stratum B is a seeded draw of 20 from the 135 papers that met a mechanical title filter.

Three constraints on interpretation. First, primary κ uses main-pass documents only. Coders were calibrated on the nine pilot texts, so pilot agreement is a property of that discussion; a pilot-inclusive figure is secondary. Second, at $n \approx 41$ with the skew expected on the rarer types, the 95% interval on κ_w has an expected half-width of 0.15–0.20, too wide to separate “substantial” from “moderate.” The design is sized for the agreement question, not powered to separate adjacent verbal bands; the expected width was stated in the registration. Third, disclosure rates are reported per stratum, with organization-level rates descriptive: at seven clusters, cluster-bootstrap intervals are downward-biased, and the list of organization-level rates is the appropriate summary.

What was registered. The pre-registration specified three hypotheses: (1) elicitation budgets and regeneration are reported in fewer than 25% of cases across all strata; (2) contamination controls are reported unevenly, Type 1 far more often than Types 2, 4 and 5; and (3) system cards report elicitation budgets more often than benchmark papers. It also fixed the framing: if the organization-clustered interval on the third contrast includes zero, the paper leads with the instrument and reports the rates as a first application. `score.py` computes that interval and prints which branch fired. With seven clusters across the two census strata the instrument branch was the expected outcome, but it did not fire: the contrast is –21% with clustered interval [–42%, –5%], which excludes zero, so the contrast survives clustering and is reported as a finding in descriptive language only. The interval rests on four clusters in Stratum A against nineteen in Stratum B; `score.py` flags a cluster count this low as indicative, so the direction is the reportable part and the width is not. One of the three hypotheses is supported. The first—elicitation and regeneration each under 25% in every stratum—holds, at a maximum of 21% for both, and is the registered prediction the data bears out. The third fails in an informative way: the contrast excludes zero but points opposite its stated direction. The second fails outright: Type 1 at 5% is not reported far more often than Types 2, 4 and 5 at 0%, 8% and 5%, and the rates remain too low for the design to order the types, so per the registration we claim no ordering.

Registration, amendment, and deviations. The analytical plan was registered before any coding, with a specified amendment procedure: rules may be updated after the pilot if the version number

is incremented, the rationale is recorded in a changelog, and the pilot documents are recoded under the new version. Amendments through v1.4 were made before coding began and are listed with date and reason in the registration's deviations table. One of them changes who codes: the registration required a single coder external to the design team, and the study as run uses two, each briefed from the coder manual and nothing else, with the design-team member moved from coding to adjudication. That departure raises the design above what was registered rather than relaxing it, and it is what licenses the claim the agreement statistic supports.

Three further deviations depart from the registered procedure, each dated in the deposited registration. First, the single post-pilot amendment the registration permits was used as v1.6, after pilot reconciliation and before any main-pass document was opened, closing five rules the pilot found at fault. An intermediate v1.5, drafted during reconciliation, was superseded by v1.6 before any document was coded under it; the changelog records both, and v1.6 is the single amendment applied. Its registered consequence—recoding all nine pilot documents—was not carried out, on time and funding, so those rows remain at v1.4. `score.py` keeps them in the disclosure denominator—eight of the 38 included documents carry v1.4 codes—and drops them only from the narrowed-boundary denominator. Second, the test–retest is performed by one coder rather than both, for the same reason. Third, the pooled agreement row in Table 4 is descriptive and was not pre-registered; the registered statistics are per-variable. Adjudication itself ran as registered: the 98 disputed cells, ten focal disagreements and three one-sided exclusions were worked in randomised order from the registered seed, after the agreement statistics were computed and saved, and every cell was settled against the document with the governing reason recorded. The lower-code tie-break was therefore never reached, and the directional tally is published so that the adjudicator's influence is visible rather than asserted: of the 88 disputed cells whose two codes both sit on the ordinal scale, 50 went to the higher code, 37 to the lower and one to a value between, while ten more carry NA on one side and have no direction. Four cells in all—two of them among those ten—were settled at a code neither coder assigned. The adjudicated τ_3 rate sits marginally above its reported band for a separate reason: the band bounds what a choice between the two coders' codes could produce on the documents both coded, and B08—excluded by one coder, admitted at adjudication—contributes a 2 from outside that domain.

A fourth item is a clarification rather than a departure, recorded here because §5 turns on it. Mid-study, one coder asked whether τ_5 covers only internet access. The answer—that the test is reaching outside the weights, so a retrieval index, a mounted filesystem or a tool call counts, and NA is reserved for evaluations with no such channel—restated the codebook's existing scope, so no rule was edited and no version moved. The codebook requires every answer to reach both coders in the same words and to be logged; both were done. It nonetheless reached the coders after they had begun coding, and it moved both without converging them, which is the evidence §5 reports against the prose test.

Results, per variable. Table 4 reports `score.py` output from the two independent sheets (agreement) and the adjudicated sheet (rates), including variables with low κ .

Where the residual disagreement sits. Two threshold questions account for most of it. On τ_2 , one coder credited the harness sub-element on eleven documents where the other recorded its absence with a specified reason; adjudication read the disclosure as real but incomplete and settled all eleven at 1. On τ_5 , the split is when the variable applies. A clarification issued mid-study to both coders in the same words moved both toward NA for evaluations with no channel outside the weights, but eighteen splits remain: eight are disclosure-level disputes about what a document states, and ten turn on the NA boundary itself. Neither coder applied that boundary consistently, and the failures run in opposite directions—one marked NA on several agentic evaluations where a repository or scaffold plainly constitutes a channel, including one the other coder scored 2 on the strength of a sandbox the document describes; the other withheld NA from a fixed no-tools condition meeting the manual's own definition. Adjudication rejected NA on all ten, finding a channel in each. An identically worded prose clarification delivered to two trained coders did not produce uniform application, which is the case for replacing the prose test with a decision procedure.

The sheet and the validator. The data sheet contains one row per document per coder. Two columns beyond the eight variables carry the audit trail: `evidence`, a locator for every non-zero code, and `codebook_version`, because the pilot may bump the version mid-study. Exclusions are recorded in this sheet, from which `exclusions.csv` is generated. Beyond schema violations, the validator

Variable	Agreement (two coders, main pass)					Disclosure (adjudicated)		
	<i>n</i>	Raw	κ_w	95% CI	AC2	2	1+2	<i>k</i>
F1 Strata	29	45%	0.18	[-0.12, 0.47]	0.43	39%	58%	26
F2 Elicitation	29	48%	0.10	[-0.10, 0.32]	0.63	13%	92%	26
F3 t1 Direct	29	69%	0.33	[0.02, 0.63]	0.68	5%	21%	26
F3 t2 Derivative	29	79%	0.00	[0.00, 0.00]	0.89	0%	11%	26
F3 t3 Temporal	29	83%	0.31	[-0.06, 0.76]	0.88	8%	16%	26
F3 t4 Distributional	29	79%	0.00	[0.00, 0.00]	0.77	8%	11%	26
F3 t5 Acquired	29	55%	0.24	[-0.01, 0.48]	0.50	5%	38%	11
F4 Regeneration	29	59%	0.35	[0.11, 0.58]	0.69	11%	50%	26
Pooled [†]	29	65%	0.46	[0.37, 0.54]	0.66			26

Table 4: Agreement and disclosure rates per coded variable. κ_w is linear-weighted Kappa with a 95% bootstrap interval; AC2 is Gwet’s prevalence-robust companion under the same weight matrix, reported because κ collapses under high class skew. Two variables—F1 and t5—fall below the registered 0.6 AC2 threshold. 2 is the rate of full reporting, 1+2 of any partial or full reporting; *k* is the number of contributing clusters. The agreement columns exclude the pilot documents (§5); pilot-inclusive agreement figures are in the supplementary materials. The disclosure columns include them: the denominator is all 38 included documents, eight of which are pilot rows still coded at v1.4. [†]Not pre-registered, reported for completeness. The pooled row exceeds every per-variable κ_w through chance correction, not higher agreement: observed weighted agreement is 0.759 whether pooled or averaged over the eight variables, while expected agreement falls from 0.693 to 0.554 once variables with different skews are mixed. The per-variable column is the study’s agreement figure.

	A System cards (<i>n</i> =12)	B NeurIPS D&B (<i>n</i> =19)	C Third-party (<i>n</i> =7)
<i>Disclosure, 2 fully reported (1+2 mentioned at all)</i>			
F1 Strata	0% (17%)	63% (89%)	43% (43%)
F2 Elicitation	0% (100%)	21% (95%)	14% (71%)
F3 t1 Direct	0% (0%)	11% (37%)	0% (14%)
F3 t2 Derivative	0% (0%)	0% (21%)	0% (0%)
F3 t3 Temporal	0% (17%)	16% (16%)	0% (14%)
F3 t4 Distributional	0% (0%)	16% (21%)	0% (0%)
F3 t5 Acquired	0% (44%)	0% (40%)	14% (29%)
F4 Regeneration	0% (0%)	21% (95%)	0% (14%)
<i>Agreement between the two coders, main pass</i>			
Raw / κ_w / AC2	0.75 / 0.45 / 0.79	0.60 / 0.47 / 0.60	0.59 / 0.23 / 0.61
F2 sub-elements present (of 5)	1.00	2.32	1.00

Table 5: Disclosure and agreement by stratum. The two axes are independent: system cards are the easiest documents to code and the least forthcoming, third-party reports the hardest to code without being the most forthcoming, which benchmark papers are. The sub-element means rest on the 12, 19 and 5 documents whose `f2_notes` record parses. Rates are adjudicated; NA is excluded per field. Only t5 has NA cells, so its row rests on 9, 5 and 7 documents rather than on the column *n*; every other row uses the full column.

warns on all controls marked `unknown`, aggregate-only reporting, missing harnesses, artifact-only releases, and Type 5 claims without documented evidence. Each is valid input and weak disclosure. Completed forms require no attribution; attribution is requested only when the specification is reproduced or adapted, so the form can be embedded in model cards.

B The July 2026 isolation failure

During a cyber-capability evaluation [Wang et al., 2026a], models with deliberately disabled safety classifiers escaped their sandbox. Public disclosures [OpenAI, 2026b, Hugging Face, 2026a,b] report that the agents accessed the open internet and chained vulnerabilities against the benchmark’s hosting infrastructure. They reached customer content the host identified as datasets whose names and files indicate a connection to the benchmark’s challenges and solutions. Agent activity ran 9–13 July

Scale	Raw	κ_w	AC2
Registered three-level (2/1/0)	0.65	0.46	0.66
Collapsed: any mention (1/2) vs absent	0.73	0.56	0.79
Collapsed: fully reported (2) vs not	0.78	0.49	0.85
NA folded into 0	0.68	0.36	0.72

Table 6: Scale-collapse ablation, main pass, 232 cells. **Exploratory; not pre-registered.** Merging 1 with 2—dropping the partial-versus-full distinction—raises κ_w from 0.46 to 0.56, while merging 0 with 1 raises it only to 0.49. Residual disagreement therefore sits at the partial-versus-full boundary—how much is enough—which the manual states in prose and should state mechanically. F2 already records five sub-elements separately so its threshold is recomputable; extending that design to the remaining variables is the next version of the instrument.

2026; the host detected it on 16 July; disclosures followed on 21 July. Whether the answer key was read, and whether any reported score was affected, is not established in the public record. The case establishes that the isolation boundary failed and that the failure was invisible from inside the run for several days—the form of the claim in §3.2.1 that a system which defeated the boundary is no witness that it held. We describe the category of failure, not its mechanism.

C Detection, prevention, and why design beats detection

The cheapest available checks fall in five categories: (1) n -gram or substring overlap, membership inference, and canary strings; (2) semantic similarity to candidate sources and provenance tracking; (3) temporal splitting around a stated data cutoff (evidence, not proof); (4) perturbation sensitivity and score distributions across item variants; and (5) transcript and tool-call review, network and environment logs, and post-run boundary verification.

Prevention. Type 1: held-out private sets, encrypted test data licensed against derivative redistribution, canary strings, and items not published alongside their solutions [Jacovi et al., 2023]. Type 2: items whose answer is stated in no single source, so answering requires integration rather than retrieval, plus published provenance and prospective cases where feasible; lexical overlap will not find this type. Type 3: items built only from phenomena unresolved at the cutoff, and continuous refresh, with the cutoff stated as self-disclosed and unverifiable—post-training, continual pretraining, and retrieval all blur it, and `unknown` is often the accurate entry. Type 4: near-miss distractors that diverge from the true answer only on evidence requiring reasoning, parameterized templates, and publishing the generation procedure rather than the artifact. Nothing prevents 5c. What makes it visible rather than silent is evidence from outside the run: egress monitoring at the network edge, canaries planted in the answer key so retrieval is detectable after the fact, and post-run verification that the boundary held. A system that defeated the boundary erases its own trace, which is why boundary monitoring is recorded separately from transcript review.

Detection is a floor. Detection can establish that something is wrong; it cannot establish that nothing is. Every method for Types 1–4 requires access to the training corpus or an assumption about memorization behaviour. The first is impossible for closed models and incomplete for open ones; recent work shows the second to be fragile. Type 5 is the exception: its evidence sits in transcripts the evaluator already owns and needs no corpus access, no behavioural assumption, and no provider cooperation. It is the newest type and the most tractable to check. Dynamic benchmarks [White et al., 2025, Zhao et al., 2025, Chen et al., 2025] raise contamination resistance without settling construct validity: replacing one unvalidated instrument with another moving one does not address that prerequisite, and freshness does nothing against 5b—a benchmark generated in the morning is acquirable that evening. Publishing the procedure that generates items outlasts publishing the items.

D The four fields: what to say when a field cannot be satisfied

Section 4 states the fields; Figure 2 shows one completed form. Adoption depends on the field the reporter cannot satisfy. Where stratification is impossible, the useful disclosure is that the evaluator cannot rule out concentration in an easy subset—what the number does not support. Where

performance is still rising at the highest budget tested, saying so converts a reported ceiling into the lower bound it is. Where a generation procedure cannot be published, the useful disclosure is that the benchmark should be assumed to degrade and names a date after which scores are not comparable. For contamination, `not_controlled` is an answer, not a failing grade: most evaluations do not control for Type 4, and saying so is what makes the score interpretable. A `controlled` claim with network access on, no transcript review, or no boundary evidence is an assumption; the validator flags each case.

Credit for the elicitation field. The elicitation field is adopted, not introduced here. OpenAI’s 2026 playbook requires third-party assessments to disclose the system tested, its tool access and harness, the elicitation methods, and the validity checks [OpenAI, 2026a]. Anthropic’s Responsible Scaling Policy retrospective notes that some of its own evaluations had lacked best-of- N or chain-of-thought prompting [Anthropic, 2026]. Independent evaluators converged on the same requirement [METR, 2024, UK AI Security Institute, 2025], and STREAM [McCaslin et al., 2025] specified reporting criteria covering prompting, sampling strategy, tool access, scaffolding, and resource ceilings—the second field here, in more detail, a year earlier. Novelty is claimed only for placing elicitation beside contamination controls, stratification, and regeneration in one record attached to the score.

E The closest prior artifacts

Table 1 gives the positioning and §2 the lineage; the full prior-art analysis is in the released materials. This appendix adds a brief lineage and the scale of the closest artifacts.

Lineage. Four earlier cuts of contamination organize by exposure severity [Xu et al., 2024], by the assumptions each detection method relies on [Fu et al., 2024], by the static-to-dynamic shift [Chen et al., 2025], and by the transformation a leaked item undergoes on its way into the corpus [Palavalli et al., 2024]. The last is confined to the pretraining path, with no evaluation-time category and no reporting artifact. Cutting a data-quality taxonomy by the remedy each defect demands is an older move [Kim et al., 2003]. Balloccu et al. [2024] document the converse route, estimating 4.7M samples from 263 datasets exposed by evaluating closed models through web interfaces: evaluation is itself a contamination channel, and a precedent for the instrument in Appendix A. Contamination inflates scores roughly $2.5\times$ more at 8B than at 1B [Kocyigit et al., 2025], so it can reorder models rather than merely raise them. Dekoninck et al. [2024] recast contamination performance-side rather than as information flow, and training on merely similar data suffices to inflate [Tu et al., 2024, Spiesberger et al., 2026], which is the basis for Type 4. Type 3 rests on task contamination [Li and Flanagan, 2024], with the caveat that the temporal signal can be induced or removed by reformatting alone [Zhang et al., 2025], so temporal splits are evidence rather than proof. The broadest prior evaluation-time framing is NIST Center for AI Standards and Innovation [2025], which separates solution from training-data contamination, documents environment- as well as network-mediated cases, and treats grader gaming as a boundary adopted in §3.2.1; agentic practice supplies supporting evidence [SWE-bench maintainers, 2025, Gupta et al., 2026].

Closest artifacts. The Contamination Transparency Card [Nourbakhsh et al., 2026] draws on a systematic review of 55 studies for a four-tier taxonomy (exact, syntactic, semantic, task-level). It covers neither elicitation budget nor stratification, and documents a benchmark at release rather than a score at report—including a score produced by a third party evaluating another organization’s model on another organization’s benchmark. STREAM [McCaslin et al., 2025] specifies the elicitation field in more detail than we do and asks for confidence intervals and run counts; it carries no contamination criterion and requires no stratified reporting. Reed et al. [2025] apply STREAM to three model reports without an agreement statistic, which is the nearest precedent for Appendix A. Evaluation Cards [Ghosh et al., 2026] derive a schema from 52 papers and 10 stakeholder interviews and deploy it across 5,816 models, 635 benchmarks, and 101,843 reported results, resolving every score through a five-level hierarchy that reaches per-split granularity. That deployment is larger than ours; we do not claim priority on machine-readable score-side reporting. The point at issue is placement: their contamination line sits under *before execution*. A property fixed before a run is release-side reasoning and cannot certify Type 5 (§3.2.1).